

SRI SIVANI COLLEGE OF ENGINEERING

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PRODUCTION PLANNING AND CONTROL

e-CONTENT

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PRODUCTION PLANNING AND CONTROL

UNIT-I

INTRODUCTION: Definition – Objectives of production Planning and Control – Functions of production planning and control – Elements of production control – Types of production – Organization of production planning and control department – Internal organization of department –Product design factors – Process Planning sheet.

UNIT-II

FORECASTING – Importance of forecasting – Types of forecasting, their uses – General principles of forecasting – Forecasting techniques– qualitative methods and quantitative methods.

UNIT-III

INVENTORY MANAGEMENT: Functions of inventories – relevant inventory costs – ABC analysis – VED analysis – EOQ model – Inventory control systems – P-Systems and Q-Systems.

Introduction to MRP & ERP, LOB (Line of Balance), JIT inventory, and Japanese concepts, Introduction to supply chain management.

UNIT-IV

ROUTING: Definition – Routing procedure – Route sheets – Bill of material – Factors affecting routing procedure. Scheduling – definition –Difference with loading.

SCHEDULING POLICIES: Techniques, Standard scheduling methods.
Line Balancing, Aggregate planning, Chase planning, Expediting, controlling aspects.

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DISPATCHING: Activities of dispatcher – Dispatching procedure –follow up – definition – Reason for existence of functions – types of follow-up, applications of computer in production planning and control.

UNIT - I

INTRODUCTION

Production Planning is a managerial function which is mainly concerned with the following important issues:

- What production facilities are required?
- How these production facilities should be laid down in the space available for production? and
- How they should be used to produce the desired products at the desired rate of production?

Broadly speaking, production planning is concerned with two main aspects: (i) routing or planning work tasks (ii) layout or spatial relationship between the resources. Production planning is dynamic in nature and always remains in fluid state as plans may have to be changed according to the changes in circumstances.

Production control is a mechanism to monitor the execution of the plans. It has several important functions:

- Making sure that production operations are started at planned places and planned times.
- Observing progress of the operations and recording it properly.
- Analyzing the recorded data with the plans and measuring the deviations.
- Taking immediate corrective actions to minimize the negative impact of deviations from the plans.
- Feeding back the recorded information to the planning section in order to improve future plans.

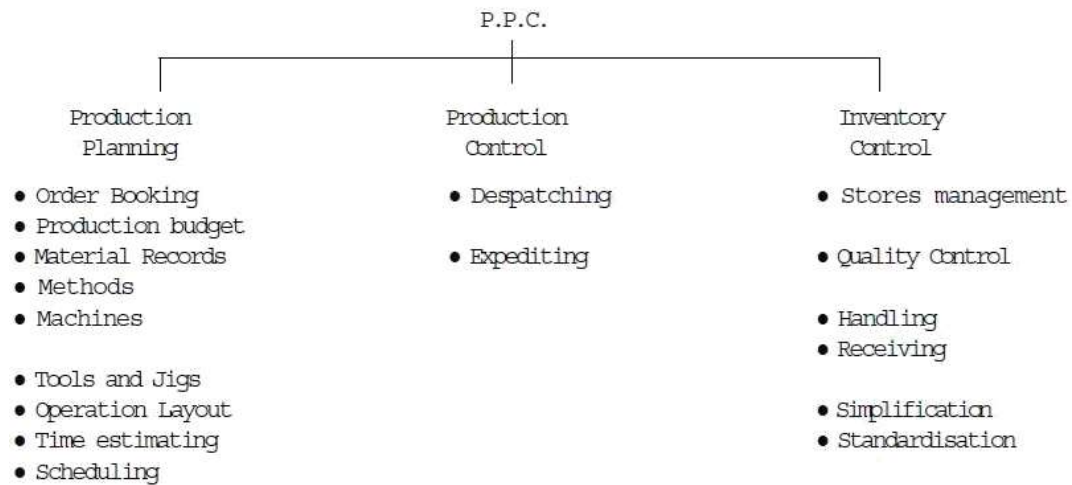
OBJECTIVES AND BENEFITS

- • Minimize costs / maximize profits
- • Maximize customer service
- • Minimize inventory investment
- • Minimize changes in production rates
- • Minimize changes in work-force levels
- • Maximize the utilization of plant and equipment

FUNCTIONS OF PRODUCTION CONTROL

- Production function encompasses the activities of procurement, allocation and utilization of resources.
- The main objective of production function is to produce the goods and services demanded by the customers in the most efficient and economical way.
- Therefore, efficient management of the production function is of utmost importance in order to achieve this objective.

ORGANISATION CHART FOR P.P.C. DEPARTMENT



An organization chart for production management department.

A block diagram depicting the architecture of a control system is shown in [Figure. _](#)

Figure 1: Architecture of Control System

Important functions covered by production planning and control (PPC) function in any manufacturing system are shown in [Table1](#) along with the issues to be covered.

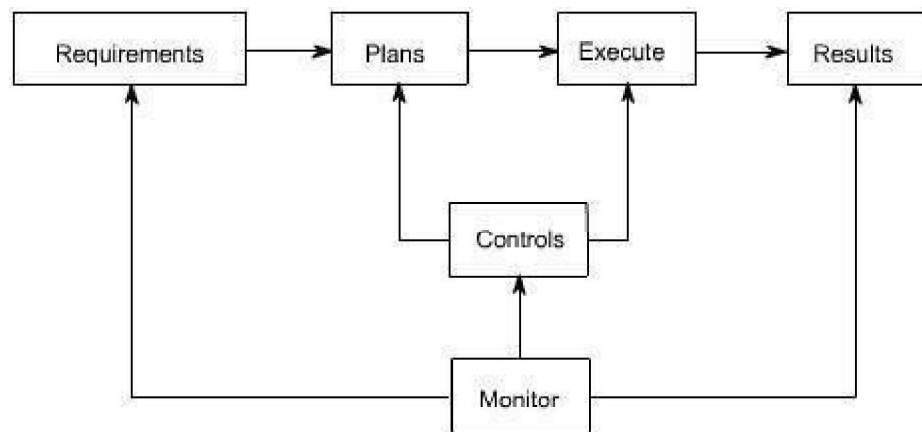


Table 1: Production Planning and control Functions

Functions	Issues to be covered
Product Design& Developme nt	Customer needs, market needs, availability of similar product, demand-supply gap, functional aspects, operational aspects, environmental aspects etc.
Demand Forecasting	Quantity, Quality, Demand pattern.
Capacity Planning	No. of machines, No. of tooling, workers, No.of flow lines, Quantity, Quality and rate of production, demand pattern.
Equipment s Selection & Maintenan cece	No. of machines, type of M/c, Quality aspects, Quantity aspects, rate of production, Cost of equipment's, support from the supplier, maintenance policy, storage of spareparts.
Tooling Selection	Compatibility between w/c steels, No. of tools, their cost, their material etc, storage policy.
Material Selection & Manageme nt	Types, specification, quality aspect, quantity aspect, cost, supplies reputation , lot size, inventory levels, setup cost, mode of transportation etc.
Process Planning	Generation of manufacture instruction, selection of M/c, tools, parameters, sequence etc.
Loading	Division of work load, assignment of tasks, uniform loading, matching between capability & capacity with job requirements.
Routing	Path selection for material movement as per the process plan and loading, minimum material handling and waiting time.
Scheduling	Time based loading, start and finish times, due dates, dispatching rules, re-scheduling.
Expediting	Operation Scheduling and order and progress reporting.

Types of Production Systems

A production system can be defined as a transformation system in which a saleable product or service is created by working upon a set of inputs. Inputs are usually in the form of men, machine, money, materials etc. Production systems are usually classified on the basis of the following:

- ☐ Type of product,
- ☐ Type of production line,
- ☐ Rate of production,
- ☐ Equipment's used etc.

They are broadly classified into three categories:

- ☐ Job shop production
- ☐ Batch production
- ☐ Mass production

Job shop Production

In this system products are made to satisfy a specific order. However that order may be produced-

- ☐ only once
- ☐ or at irregular time intervals as and when new order arrives
- ☐ or at regular time intervals to satisfy a continuous demand

The following are the important characteristics of job shop type production system:

- ☐ Machines and methods employed should be general purpose as product changes are quite frequent.
- ☐ Planning and control system should be flexible enough to deal with the frequent changes in product requirements.
- ☐ Man power should be skilled enough to deal with changing work conditions.
- ☐ Schedules are actually nonexistent in this system as no definite data is available on the product.
- ☐ In process inventory will usually be high as accurate plans and schedules do not exist.
- ☐ Product cost is normally high because of high material and labor costs.
- ☐ Grouping of machines is done on functional basis (i.e. as lathe section, milling section etc.)
- ☐ This system is very flexible as management has to manufacture varying product types.
- ☐ Material handling systems are also flexible to meet changing product requirements.

Batch Production

Batch production is the manufacture of a number of identical articles either to meet a specific order or to meet a continuous demand. Batch can be manufactured either-

- ☐ only once
- ☐ or repeatedly at irregular time intervals as and when demand arise
- ☐ or repeatedly at regular time intervals to satisfy a continuous demand

The following are the important characteristics of batch type production system:

- ☐ As final product is somewhat standard and manufactured in batches, economy of scale can be availed to some extent.
- ☐ Machines are grouped on functional basis similar to the job shop manufacturing.
- ☐ Semi-automatic, special purpose automatic machines are generally used to take advantage of the similarity among the products.
- ☐ Labor should be skilled enough to work upon different product batches.
- ☐ In process inventory is usually high owing to the type of layout and material handling policies adopted.
- ☐ Semi-automatic material handling systems are most appropriate in conjunction with the semi-automatic machines.

Mass Production

In mass production, same type of product is manufactured to meet the continuous demand of the product. Usually demand of the product is very high and market is going to sustain same demand for sufficiently long time.

The following are the important characteristics of mass production system:

- As same product is manufactured for sufficiently long time, machines can be laid down in order of processing sequence. Product type layout is most appropriate for mass production system.
- Standard methods and machines are used during part manufacture.
- Most of the equipment's are semi-automatic or automatic in nature. Material handling is also automatic (such as conveyors).
- Semi-skilled workers are normally employed as most of the facilities are automatic.
- As product flows along a pre-defined line, planning and control of the system is much easier. Cost of production is low owing to the high rate of production.
- In process inventories are low as production scheduling is simple and can be implemented with ease.

PRODUCT DESIGN

- Product design is a strategic decision as the image and profit earning capacity of a small firm depends largely on product design. Once the product to be produced is decided by the entrepreneur the next step is to prepare its design. Product design consists of form and function. The form designing includes decisions regarding its shape, size, color and appearance of the product. The functional design involves the working conditions of the product. Once a product is designed, it prevails for a long time therefore various factors are to be considered before designing it. These

factors are listed below: -

- (a) Standardization
- (b) Reliability
- (c) Maintainability
- (d) Servicing
- (e) Reproducibility
- (f) Sustainability
- (g) Product simplification
- (h) Quality Commensuration with cost
- (i) Product value
- (j) Consumer quality
- (k) Needs and tastes of consumers.

Above all, the product design should be dictated by the market demand. It is an important decision and therefore the entrepreneur should pay due effort, time, energy and attention in order to get the best results.

TYPES OF PRODUCTION

Broadly one can think of three types of production systems which are mentioned here under: -

- (a) Continuous production
- (b) Job or unit production
- (c) Intermittent production

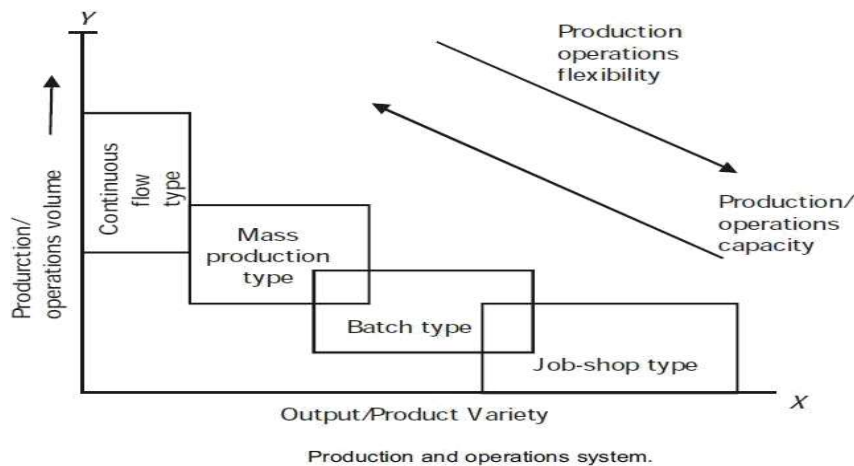
(a) **Continuous production:** - It refers to the production of standardized products with a standard set of process and operation sequence in anticipation of demand. It is also known as mass flow production or assembly line production. This system ensures less work in process inventory and high product quality but involves large investment in machinery and equipment. The system is suitable in plants involving large volume and small variety of output e.g. oil refineries reform cement manufacturing etc.

(b) **Job or Unit production:** - It involves production as per customer's specification each batch or order consists of a small lot of identical products and is different from other batches. The system requires comparatively smaller investment in machines and equipment. It is flexible and can be adapted to changes in product design and order size without much inconvenience. This system is most suitable where heterogeneous products are produced against specific orders.

(c) **Intermittent Production:** Under this system the goods are produced partly for inventory and partly for customer's orders. E.g. components are made for inventory but they are combined differently for different customers. . Automobile plants, printing presses, electrical goods plant are examples of this type of manufacturing.

Intermittent production

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- Automobile plants, printing presses, electrical goods plant



MARKETING ASPECT

- Sales and Marketing is a key function whose participation is often hard to enlist.
- Sales and Marketing are critical functions in this process, since they provide the starting point of the planning and scheduling process -- the forecasts and customer order demands
- They are also vital from the viewpoint of providing the proper customer perspective whenever changes need to be made to plans and schedules based on mismatches of resources to customer demands.
- Only with a proper level of participation in Planning and Scheduling, can Sales and Marketing optimally leverage its performance and create a trusting and consensus based working relationship with Manufacturing, Purchasing, Planning, Engineering and all other functions in the company.

Standardization

- Sizes for screws, nuts bolts and other threaded fasteners were first standardized based on work of by Joseph Whitworth..
- Pipe sizes
- Shoe size standardization
- The screw base size and thread dimensions of electric lamp bulbs was standardized by Thomas Edison.
- Electrical voltage and frequency
- Electrical wiring and device standards